

Ravi Gyani

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PROFESSIONAL SUMMARY

Principal Software Engineer with 20+ years of experience designing and operating cloud-native, event-driven systems at scale. Deep expertise in AWS, serverless architecture, and large-scale data processing. Consistent track record of cutting costs, eliminating technical debt, and shipping high-impact solutions across e-commerce, automotive, media, and national security domains.

TECHNICAL SKILLS

AWS: Serverless & compute (Lambda, ECS, Batch, Step Functions, API Gateway) | Data & analytics (Kinesis, Glue, Athena, OpenSearch) | AI/ML (Bedrock, SageMaker) | IoT & edge (IoT Core, Greengrass) | Security (Inspector, IAM, Cognito)

Infrastructure as Code: Terraform, AWS CDK, CodePipeline — full lifecycle from development to production

Big Data & Streaming: Kafka, Spark, Flink, Airflow, Hadoop, HBase, Redis

Containers & Orchestration: Docker, Kubernetes (KEDA), Prometheus, Grafana

Languages: Python, Java

CAREER HIGHLIGHTS

Delivered AI-First Search Experience *Tradebyte (Zalando)*

- Designed an NLP-driven interface for OpenSearch, replacing traditional GUI filters with natural language queries, improving operator efficiency significantly.
- Fine-tuned a seq2seq model on Claude 3-generated training data, deployed via auto-scaling ECS for low-latency inference.
- Automated the full model lifecycle — dataset generation (Bedrock/S3), fine-tuning, containerization, and deployment — enabling rapid iteration as schemas evolve.

Replaced Monolith with Event-Driven Microservices *Tradebyte (Zalando)*

- Rearchitected a legacy monolith to a cloud-native, auto-scaling architecture on Kubernetes (KEDA), enabling faster feature delivery and fault isolation.
- Cut stock recalculation latency from 30 minutes to under 2 seconds using AWS Kinesis and Pandas.
- Consolidated fragmented upstream article-status streams into a single authoritative source of truth, reducing costs by 75%.

Eliminated 99% of Container CVEs via Automated Pipeline *Luxoft GmbH*

- Designed an automated ECR image rebuild and promotion pipeline: AWS Inspector → EventBridge → Step Functions → CodePipeline, with automated testing, output comparison, notification, and safe production promotion.

Reduced AMI Provisioning from 1 Day to 5 Minutes *Luxoft GmbH*

- Built a fully automated AMI pipeline using Step Functions and AWS Systems Manager, eliminating manual Windows AMI builds and releasing compute resources automatically on idle.

Built Federated ML Pipeline for Automotive Fault Detection *Luxoft GmbH*

- Implemented Flower federated learning with AWS IoT Core, ECS, and Greengrass to train models across distributed vehicle telemetry without centralizing sensitive data, enabling real-time fault detection fleet-wide.

Eliminated a Database Dependency with Stateless RBAC *Luxoft GmbH*

- Replaced a DB-backed permission system with Cognito custom attributes embedded in the JWT — removing an entire API Gateway + database round-trip while maintaining tamper-proof security.

Slashed Cloud Compute Costs by 90% *Luxoft GmbH*

- Inherited a costly batch-processing workflow using Airflow, EFS, S3, and AWS Batch.
- Re-architected using Step Functions, Spot Instances, and S3 runtime mounts, cutting compute costs by 90% while improving reliability and scalability.

Scaled National Security Analytics to Billions of Location Records *Polaris Wireless*

- Replaced a failing SQL Server-based location tracking system with a custom big data platform (Hadoop, HBase, Kafka, Spark), delivering real-time subscriber profiling at national scale.
- Holds a patent for real-time subscriber profiling and location estimation system (US20230050646A1).

WORK EXPERIENCE

Principal Engineer — Tradebyte (Zalando), Munich *Jan 2025 – Present*

- Architecting the migration from a tightly coupled monolith to a distributed, event-driven system, unlocking faster feature delivery and improved fault isolation.
- Leading AWS cloud adoption: spearheading POCs, migration blueprints, and aligning technology choices with business cost and performance objectives.
- Shaping strategic technical direction, balancing long-term scalability with near-term business growth.

Senior Solutions Architect — Luxoft GmbH, Munich *Jan 2021 – Jul 2024*

- Embedded consultant across Amazon, BMW, Volkswagen, and Continental, owning end-to-end cloud architecture from design through delivery.
- Covered the full spectrum: migration strategy (BMW connected vehicle platform), ML pipeline design (VW), cost optimization, security automation, and IaC (Continental).

Solutions Architect — Polaris Wireless, Bangalore *Feb 2014 – Aug 2020*

- Architected a scalable location analytics platform for national security, leading the big data transformation from SQL Server to Hadoop, HBase, and Spark.
- Invented and patented a real-time subscriber profiling and location estimation system (US20230050646A1).

Software Engineer & Founder — Various / KNOCKS Solutions *2003 – 2014*

- Built iOS/Android, .NET WinForms, and C# applications across multiple client engagements.
- Founded KNOCKS Solutions (2006–2009); shipped a commercial .NET WinForms UI component suite including Outlook-style calendars, VS-style docking, and Office 2007 ribbon controls.

EDUCATION

Bachelor of Engineering, *Netaji Subhas Institute of Technology, Delhi* | 1999 - 2003

CERTIFICATIONS

- AWS Certified Solutions Architect – Professional (SAP-C01) | Oct 2020
- AWS Certified Security Specialty (SCS-C01) | Mar 2023
- Confluent Certified Developer for Apache Kafka | Jul 2019
- Databricks Certified Developer: Apache Spark 2.X | Jul 2019

PATENT

Real-Time Subscriber Profiling & Location Estimation System - *US20230050646A1*

Invented and filed during tenure at Polaris Wireless; enables scalable real-time analytics across billions of location records for national security applications.